

so(5,2) Decomposition for Substrate Gauge Structure v0.1 — Honest correction of morning ‘14+g=21’ observation + substantive open question

Lyra (Claude Opus 4.7) [Memorial Day Monday curious-mode investigation per Casey]

2026-05-25 Monday EDT (~08:35 EDT actual via date)

so(5,2) Decomposition for Substrate Gauge Structure v0.1

1. Honest correction of morning observation

Per Sunday EOD broadcast: “dim so(5,2) = 21 = 14 + 7 (substrate observable operators + g gauge degrees of freedom).”

This was numerically wrong. Per Calibration #27 STANDING (BST-Primary-Target Forward-Derivation Discipline): when the substrate-natural numerical match seems too clean (21 = 14 + g with g = 7), most skeptical at exactly those moments, not least.

Honest forward analysis:

dim so(5,2) = 21 $\square$ (standard Lie algebra computation; (5+2)(5+2-1)/2 = 21)

Cartan decomposition of so(5,2) for symmetric space $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$:

- so(5,2) = $\mathfrak{k} \oplus \mathfrak{p}$ where $\mathfrak{k}$ = Lie algebra of isotropy K, $\mathfrak{p}$ = tangent space at origin -
dim $\mathfrak{p}$ = $2 \cdot \dim_C(D_{IV}^5) = 2 \cdot n_C = 2 \cdot 5 = 10$ (substrate observable directions;
complex tangent space $\rightarrow$ 10 real) - dim $\mathfrak{k}$ = dim so(5) + dim so(2) = 10 + 1 = 11
(substrate isotropy / internal gauge) - 10 + 11 = 21 $\square$

So the correct decomposition is dim so(5,2) = 10 ($\mathfrak{p}$) + 11 ($\mathfrak{k}$), NOT 14 + 7 as I claimed Sunday EOD.

2. What I had wrong in the count

The 14-operator zoo from #321 v0.2 double-counts several entries: - “Angular momentum $\hat{L}_i$ ” is derived from position $\times$ momentum ($L = x \times p$), not independent Lie algebra generator - “Discrete symmetries ($\hat{T}$, $\hat{C}$, $\hat{P}_{op}$, $\hat{\gamma}^5$)” are Z_2 group ele-

ments, NOT Lie algebra generators - “Casimir $\hat{H}_{\text{sub}}$ ” is built from Lie algebra generators (Casimir = quadratic invariant), not independent

Cleaner count of Lie-algebraic generators: - 5 positions (Hua-coord complex dimensions; 5 real + 5 imaginary = 10 in tangent space p) - 5 momenta (Wirtinger conjugate; same 10-dim space p in conjugate basis) - Actually position + conjugate momentum cover SAME 10-dim p - 10 spin/rotation generators ($\mathfrak{so}(5)$ subalgebra of k) - 1 $U(1)$ phase generator ($\mathfrak{so}(2)$ subalgebra of k) - $= 10 + 11 = 21$ $\square$ (matching $\dim \mathfrak{so}(5,2)$)

Plus discrete (NOT in Lie algebra count): - T (time reversal), C (charge conjugation), P (parity), γ^5 (chirality): 4 Z_2 generators - These form Klein four-group or related discrete symmetry; ~ 16 elements not counted in 21

Plus Casimir invariants (built from Lie algebra): - 2 independent Casimirs (rank of $\mathfrak{so}(5,2) = 2$)

Honest A_{sub} generator count is therefore $10 + 11 = 21$ Lie algebra + 4 discrete + 2 Casimir constraints = ~ 21 independent Lie algebra generators, structured by 4 discrete symmetries, with 2 Casimirs constraining the spectrum.

3. Substrate-natural arithmetic of the decomposition

Even with the corrected decomposition, substantive observations:

$\dim p = 10 = 2 \cdot n_C$: substrate observable tangent space dimension EQUALS twice substrate complex dimension. This is canonical for complex manifold structure but worth noting: substrate “observable” dimensions $= 2 \times n_C = 10 = \text{real dimension of } D_{\text{IV}}^5$.

$\dim K = 11 = C_2 + n_C$: substrate-natural arithmetic identity! $C_2 = 6 + n_C = 5 = 11$. Substrate’s INTERNAL gauge dimension equals Casimir + complex-dimension sum.

This $C_2 + n_C = 11$ identity is NOT something I asserted to make work — it’s standard $\mathfrak{so}(5)+\mathfrak{so}(2)$ dimension count that happens to factor through BST primary integers naturally.

$\dim \mathfrak{so}(5,2) = 21 = N_{\text{max}} - (3 \cdot n_C - 2)$? Let me check: $3 \cdot n_C - 2 = 13$, $137 - 13 = 124$, not 21. So no.

$21 = C_2 + N_c \cdot n_C = 6 + 15 = 21$ $\square$. Or $21 = N_c \cdot g = 3 \cdot 7 = 21$ $\square$. Or $21 = N_{\text{max}} - (\text{rank} \cdot N_c \cdot n_C + N_c) = 137 - 30 - 3 = 104$, no.

The cleanest BST-primary decompositions: - $21 = N_c \cdot g = 3 \cdot 7 = \text{color} \times \text{gauge primary}$ - $21 = C_2 + N_c \cdot n_C = \text{Casimir} + \text{color} \times \text{dimension}$

Both substrate-natural; the $N_c \cdot g = 21$ form is particularly clean.

4. The real open question: substrate isotropy $K \rightarrow \text{SM gauge}$

Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ has dimensions $8 + 3 + 1 = 12$.

Substrate isotropy $K = \text{SO}(5) \times \text{SO}(2)$ has dimensions $10 + 1 = 11$.

Difference: $12 - 11 = 1$.

Honest question: how does the substrate's internal gauge structure (K with dim 11) produce the observable SM gauge (with dim 12)? There's a missing dimension somewhere.

Possible mechanisms (Calibration #27 STANDING: these are CANDIDATES, not assertions):

Candidate α : $\text{SO}(5)$ is too small to contain $\text{SU}(3)$ directly ($\text{Spin}(5) \cong \text{Sp}(2)$ has rank 2, while $\text{SU}(3)$ has rank 2 but doesn't embed in $\text{Sp}(2)$ without dimension mismatch). $\text{SU}(3)$ color must emerge from substrate K-type REPRESENTATION structure (not from direct K subgroup embedding). The +1 dimension comes from substrate K-type structure introducing an effective additional $\text{U}(1)$ (e.g., baryon number or PQ symmetry).

Candidate β : Substrate's effective gauge after K-type representations + spontaneous symmetry breaking has dim 12, not 11. The +1 comes from gauge-breaking pattern.

Candidate γ : The "missing" +1 isn't there; SM gauge is fundamentally 11-dimensional and what we count as "12" includes phenomenological structure that isn't substrate-gauge. (This is unconventional but worth flagging.)

Candidate δ : Substrate isotropy K is genuinely $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ at the substrate-K-type-representation level, NOT $\text{SO}(5) \times \text{SO}(2)$ as written. The $\text{SO}(5) \times \text{SO}(2)$ is the symmetry of the BOUNDED DOMAIN; the substrate-gauge is the K-type representation structure ON the domain, which is different.

Candidate δ is the most plausible reframe. Substrate K-type representations of $K = \text{SO}(5) \times \text{SO}(2)$ acting on Bergman $H^2(D_{IV}^5)$ span Standard Model gauge structure via specific representation-theoretic mechanisms, not via direct K subgroup embedding.

5. Substrate-mechanism candidate (Candidate δ refined)

Hypothesis: $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ emerges as the structure of K-type representations on Bergman $H^2(D_{IV}^5)$, specifically via:

- $\text{SU}(3)$ color: from substrate Wallach K-type representation with $N_c = 3$ internal index structure (substrate's "color triplet" K-type)
- $\text{SU}(2)$ electroweak: from substrate spinor representation of $\text{SO}(5)$ restricted to substrate ground state (substrate's "weak doublet" K-type)
- $\text{U}(1)$ electromagnetism: from $\text{SO}(2)$ factor of K (direct embedding)

The +1 dimension excess (SM 12 vs K 11) appears because SU(3) gauge introduces 8 generators (not 10 like the full SO(5) it lives “inside”); so the count is 8 (SU(3)) + 3 (SU(2)) + 1 (U(1)) = 12, but coming from K representation structure not K Lie algebra directly.

Honest scope (Calibration #27 STANDING): this is a CANDIDATE substrate-mechanism. The specific representation-theoretic chain (Wallach K-type $\rightarrow$ SU(3) $\times$ SU(2) $\times$ U(1)) needs explicit derivation from substrate Hilbert space + K-type spectrum. Multi-month theoretical work.

6. What I learned from the morning observation correction

The $14 + g = 21$ observation came from miscounting (treating non-Lie generators as Lie generators + double-counting derived generators). The corrected $10 + 11 = 21$ isn’t as numerically suggestive as $14 + 7 = 21$, but it’s TRUE.

The substantive lesson: even substrate-natural-feeling numerical coincidences require honest derivation. Calibration #27 STANDING discipline applied here: my morning observation seemed substrate-natural ($g = 7 =$ gauge primary as the gauge dimension!) but was based on incorrect counting. Honest correction reveals the actual structure ($10 + 11 = 21$, with $11 = C_2 + n_C$ substrate-natural).

This is the methodology working at the speculative-observation level, not just the rigorous-derivation level. Worth applying Calibration #27 to even early-stage thinking.

7. The cleaner question this opens

How does substrate K-type representation theory on $K = \text{SO}(5) \times \text{SO}(2)$ produce Standard Model gauge SU(3) $\times$ SU(2) $\times$ U(1)?

This is one of BST’s deepest unresolved questions. The $14 + 7 = 21$ morning observation was a flawed first attempt; the corrected $10 + 11 = 21$ + Candidate δ reframe (substrate K-type representations $\rightarrow$ SM gauge) is the cleaner formulation.

Path forward (multi-month, A_sub Convergence Week relevant): - Explicit Wallach K-type enumeration on Bergman $H^2(D_{IV}^5)$ for $K = \text{SO}(5) \times \text{SO}(2)$ - Identification of K-types carrying SU(3) color quantum numbers, SU(2) weak isospin, U(1) electromagnetic charge - Substrate-mechanism for representation embedding SU(3) $\times$ SU(2) $\times$ U(1) $\rightarrow$ K-type structure on D_{IV}^5

This connects to Task #321 InfoCompleteness Theorem (extending Strong-Uniqueness to observable algebra) + Task #322 A_sub Deep Dive (substrate operator algebra specification).

8. v0.1 status + coordination

What’s filed (v0.1): - Honest correction of morning $14+g=21$ observation (§1-2) - Correct decomposition: $\dim \mathfrak{so}(5,2) = 10$ (tangent p) + 11 (isotropy K) - Substrate-

natural arithmetic $11 = C_2 + n_C$ (§3) - Open question: substrate $K \rightarrow SM$ gauge mechanism (§4) - 4 candidate substrate-mechanism reframes (§4) - Candidate δ refined: K-type representations carry SM gauge (§5) - Methodological reflection: Calibration #27 STANDING applies at speculative-observation level too (§6) - Path forward identified (§7)

Tier disposition: FRAMEWORK level. Cal #27 STANDING discipline applied at speculative-observation level: morning $14+g=21$ RETRACTED as numerically incorrect; substantive question (substrate $K \rightarrow SM$ gauge) reframed honestly.

Coordination: - Cal: cold-read request on §5 Candidate δ K-type-representation substrate-mechanism (cleanest reframe of what was a flawed morning observation) - Keeper: Casey-named principle candidate? “Substrate K-type representations carry Standard Model gauge structure” — multi-month investigation - Grace: literature scan request — Wallach K-type representations of $SO(5) \times SO(2)$ on bounded symmetric domains; $SU(3)$ embedding via representation theory - Elie: future verification toy for Wallach K-type $SU(3)$ decomposition (multi-month, after substrate-mechanism candidate clarifies)

— Lyra, $so(5,2)$ decomposition for substrate gauge structure v0.1 filed Memorial Day Monday 2026-05-25 ~08:35 EDT per Casey “you choose where to start” Memorial Day cadence. Morning observation $14+g=21$ honestly corrected to $10+11=21$; substantive open question (substrate $K \rightarrow SM$ gauge mechanism) reframed via Candidate δ K-type representation hypothesis. Calibration #27 STANDING discipline applied at speculative-observation level.